

COURSE TITLE : DIGITAL CIRCUITS LAB
COURSE CODE : 3217
COURSE CATEGORY : B
PERIODS/WEEK : 5
PERIODS/SEMESTER: 75
CREDITS : 3

On completion of this course the student will be able to

1. Familiarization of
 - (i) TTL and CMOS Logic IC's by Verification of Truth Tables.
 - (ii) Universal gates for implementing other logic functions.
2. Design & verify binary to gray and gray to binary converter.
3. Design, setup and verify the
 - (i) Half and full adder circuits.
 - (ii) Half and full subtractor circuits.
4. Design of combinational logic circuits from Boolean equation.
5. Set up Four bit adder and subtractor using ICs 7483.
6. Design BCD adder using IC 7483.
7. Design BCD to Seven Segment Decoder using 7447/7446 and Display.
8. Design 4 to 1 MUX using NAND Gates.
9. Implement 8 to 1 MUX using the multiplexer IC 74151.
10. Implement combinational logic using multiplexer ICs.
11. Verify the truth table RS, D, JK, M/S JK, T Flip flops using NAND gate.
12. Design and construct asynchronous mod- N counter using flip-flops.
13. Design and construct synchronous mod - N counters using flip-flops.
14. Design and construct 3-bit up/down counter.
15. Design and construct 4 bit shift register using flip flops.
16. Implement the operation of shift register ICs.
17. Construct Johnson counter and Ring counter.